

Anyone else worried about the energy + human dependency side of AI?

Posted by u/NoLawfulness3621 • 0 points • 21 comments

Hey, this might sound a little weird but I can't shake the thought, so I figured I'd throw it out here and see if anyone else has been thinking the same.

A lot of the AI conversation I see is about jobs being replaced or robots taking over, but that's not really what's on my mind. What gets me is the energy and sustainability side of all this.

These models need massive data centers, which eat up crazy amounts of electricity and water just to run and cool. Every company is already talking about building even bigger ones. On top of that, people are getting more and more attached to AI for even basic stuff — writing, planning, answering questions, entertainment, whatever. The dependency is real.

So here's my thought:

What happens if, down the line, we realize we just can't sustain both humans and AI at the same level? Like, energy grids, food production, climate change, all fighting for the same resources — and AI's demand keeps climbing.

At some point, do governments or societies say, "this isn't sustainable, shut some of it off"? And if that happens, what's the impact on us? We're already getting used to leaning on it for so many little things that without it, people might struggle to function — almost like a "zombie mode" because we forgot how to operate without the crutch.

I don't know, maybe it's just a random thought spiraling in my head, but it feels like no one is talking about it. Everyone's hyped about new models and features, but is anyone seriously weighing whether we can actually support the infrastructure long-term?

Curious if I'm overthinking this or if others are seeing the same risk. Would love to hear perspectives.

Comments

u/DescriptionSevere335 • 1 points • 2025-09-17 10:56 UTC

I get your worry, but luckily, reports from the last weeks show great news, AI uses way less power than we thought, like 10x less. It also uses way less water than we thought.

This is partly because we were working with estimates at the start, now have real data. But also because of massive improvements in the hardware and how the models are made, leading to them being more efficient. This trend will only continue (exponentially).

I've been worrying about climate change and sustainability, like panicked, for the last 15-20 years. I kind of assumed we were doomed; as we would never for the good of humanity take the action that was needed. It involves billions of people volunteering to give things up. That's not going to happen. We can't agree even it's a problem. Or what's the best way to solve the problem.

BUT! Economic stimulants work better. Elon Musk has gone insane, but he also quickly made electric vehicles outnumber vehicles that run on fossil fuels. This works. And we are seeing happen with energy. A year and half ago, OpenAI said they want to buy and build 25 nuclear power plants. That's half the number of active ones in the US. Just for AI. It is happening because there is money to be made. And lots of it. It's driving innovation and a transition to nuclear, which is the safest most sustainable power source we have access to.

I think AI is directly causing us to increase the amount of energy we can harness and use exponentially. This exponential increase in energy will let us do wonderful things to the climate in the end.

u/NoLawfulness3621 • 1 points • 2025-09-17 12:33 UTC

I get where you're coming from — and you're right, AI has gotten way more efficient thanks to better chips, smarter training, and cooling tech. But the story isn't as simple as "10× less power and water."

For context:

In 2024, data centers worldwide used about 415 terawatt-hours of electricity (≈1.5% of all global electricity).

By 2030, the International Energy Agency expects that number to more than double to around 945 TWh, mostly because of AI workloads.

On water: some data centers do use more efficient cooling now, but as AI demand scales, total water use still rises — and it's already caused tension in some communities.

So yes, per-query or per-training efficiency is improving, but the overall footprint is still growing fast.

Where you're absolutely spot-on is the economics: companies are chasing efficiency and clean energy because it saves money and opens up massive business opportunities.

That's why you're seeing real moves toward nuclear and other sustainable energy sources.

In short: AI is getting cleaner per task, but the demand explosion means its total energy and water impact is still climbing. The upside is that the profit motive is pushing the whole system toward cleaner power a lot faster than government policy alone ever could.

u/DescriptionSevere335 • 1 points • 2025-09-17 12:41 UTC

Typically, the more we use something, the better we get at making it. We make it more efficient, but still better. Less inputs needed, less waste created.

But yes, AI improvements + this will still lead to more water and energy being used than was being used. The energy will NEED to be sustainable though. We can't power AI with coal and gas. It's impossible, at the scale we want to do it on. This leads to a bigger demand than there is supply.

I would normally be scared about that. But there is such a crazy scramble to be the big winner and get to AGI first. This will make it financially viable to speed up and intensify the energy transition. No idea if we will get there. But AI's energy needs + the greed of some people means me having hope for the world in the first time in a decade. I was sure we were doomed...now I see a chance that AI's energy needs save us.

u/Scary_Historian_8746 • 2 points • 2025-09-15 13:43 UTC

You're not overthinking it at all. The energy footprint of big AI models is massive, and most of the hype skips over that fact. If we scale endlessly without addressing sustainability, it's not just an economic

problem, it becomes an environmental one too. Dependency makes it worse, because pulling the plug later would hit society like a shock

u/NoLawfulness3621 • 1 points • 2025-09-16 13:51 UTC

So true

u/GuyThompson_ • 1 points • 2025-09-13 01:13 UTC

It's going to be bonkers. I'm already deep into energy ETFs and they are going absurdly well. AI is eating all the energy it can get

u/NoLawfulness3621 • 1 points • 2025-09-16 13:55 UTC



u/Slight_Republic_4242 • 2 points • 2025-09-12 20:38 UTC

no not all i am using ai voice agent for automation of my real estate sales calls using dograh ai offers human like conversation reduces the workload pressure on sales team .. not dependency but helping hand

u/ElectroNetty • 2 points • 2025-09-12 19:48 UTC

AI is already being used to advance other technologies, especially fusion. The energy and cooling problem will be solved over the coming decades.

The next part about becoming an idiocracy is true, but not because of AI. Education is lacking everywhere on Earth and keeping the majority of people stupid is more profitable so it's a given that society will go that way while greed is the driving factor.

u/WhaleFactory • 2 points • 2025-09-12 19:10 UTC

I'm more worried about the social fabric being torn to shreds on its own right now.

I do think people should learn to run local models as they are very capable now. AI + Big Tech is going to be a very dangerous thing unless it is regulated appropriately.

u/Many_Community_3210 • 2 points • 2025-09-12 18:04 UTC

If society needs more energy then more energy will be produced. environmental hatred of nuclear is greatly misplaced and our energy generating capacity will increase due to demand. Will this usher the environmental collapse some have been promising since I can remember? I think not.

u/reddit455 • 3 points • 2025-09-12 17:18 UTC

>but is anyone seriously weighing whether we can actually support the infrastructure long-term?

that infrastructure can be built.

****Google to fund development of three nuclear power sites****

<https://www.world-nuclear-news.org/articles/google-to-fund-elementl-to-prepare-three-nuclear-power-sites>

****Meta becomes the latest big tech company turning to nuclear power for AI needs****

<https://apnews.com/article/meta-facebook-constellation-energy-nuclear-ai-a2d5f60ee0ca9f44c183c58d1c05337c>

****Microsoft describes Three Mile Island plant as a once-in-a-lifetime opportunity****

<https://penncapital-star.com/economy/microsoft-describes-three-mile-island-plant-as-a-once-in-a-lifetime-opportunity/>

> Curious if I'm overthinking this or if others are seeing the same risk

the problem is not "where to get energy" - the problem is BUILDING IT. it's a matter of WILL.

*****there's no technical barrier.*****

solar is more than enough to run MOST things outside of large industry and AI data centers.

[<https://thatsbyers.com/blog/how-much-energy-does-the-sun-really-provide/>]
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According to the U.S. Department of Energy, 430 quintillion Joules of energy from the sun hits the earth each hour; humans use 410 quintillion Joules a year, and the average American household uses about 40 billion Joules of electricity.

u/NoLawfulness3621 • 1 points • 2025-09-12 18:36 UTC

Thanks for this

u/v01dm4n • 2 points • 2025-09-12 17:12 UTC

Similar things have been said about cars, planes, computers.

There were those who were worried if we need that much electricity and whether it's sustainable.

Then those who questioned the furnaces.

Then the furniture, the housing, and even fire.

Humans have been taming nature since forever for their own benefit and in each generation there have been some who become the victims, some who exploit and some who refuse to participate in the circus.

Unfortunately I see no stopping to this until we hit limits.

u/NoLawfulness3621 • 1 points • 2025-09-13 01:12 UTC

Well said

u/sludge_monster • 2 points • 2025-09-12 17:09 UTC

Whatever you do, don't look into how concrete is made.

u/Muted_Hat_7563 • 3 points • 2025-09-12 17:05 UTC

The earth is resilient and can sustain FAR more growth than people think it can. HOWEVER... these are valid concerns and resource constraints will definitely become a problem, but with ai, especially agi/asi, we can improve on everything and make everything from food production, to housing, energy production with fusion, and everything else much more orders of magnitude more efficient and sustainable. The future will not be as grim as you think it may become, because you are thinking in today's terms, and not what will happen as a result of the tremendous amount of scientific discoveries and advances that can be had with ai.

u/NoLawfulness3621 • 1 points • 2025-09-12 20:16 UTC

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